

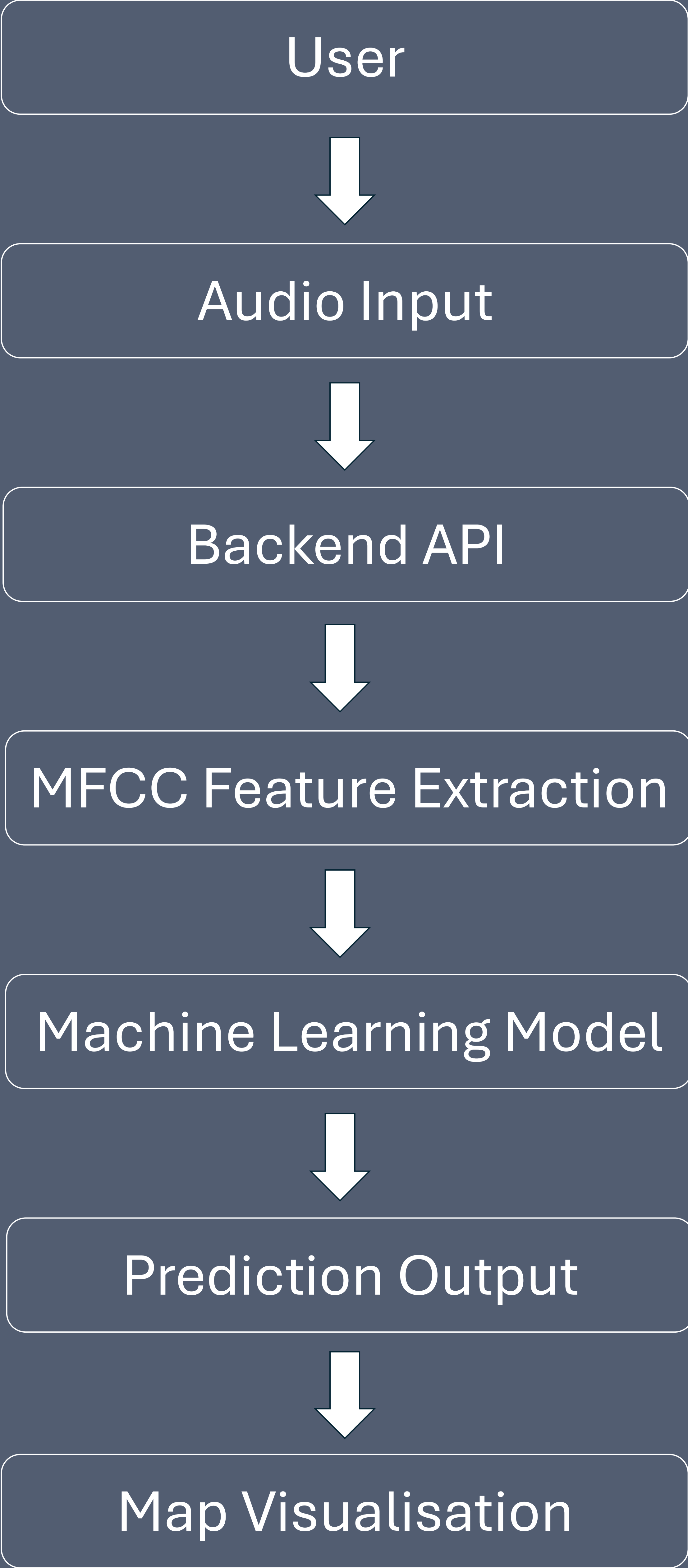
# Mapping the Spoken Word:

## A Geo-Spatial Machine Learning Approach to Irish Accent Recognition

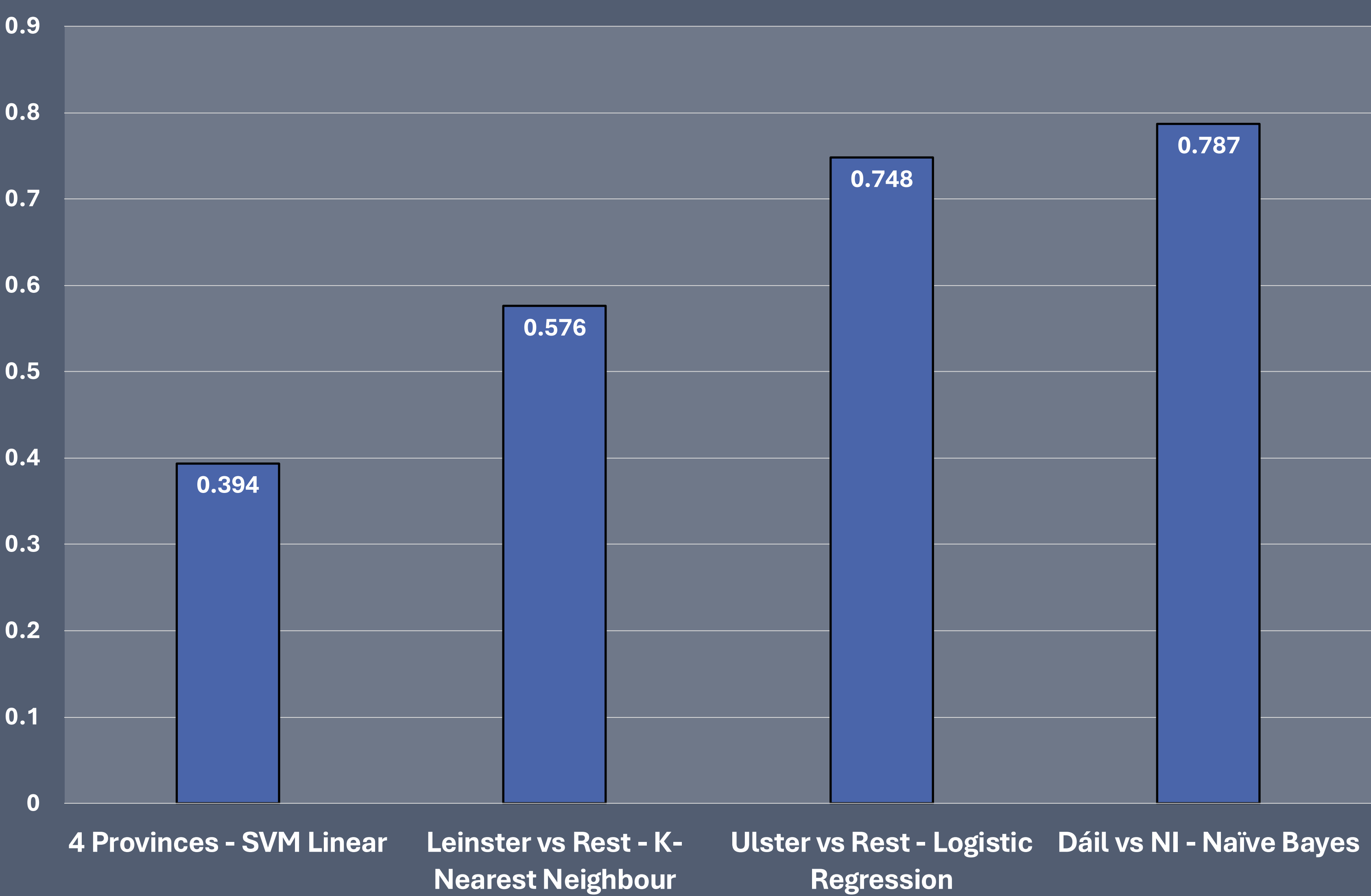
Cianán Finn, TU856, TU Dublin



### System Diagram



Balanced Accuracy of Different Models on Different Data



#### Introduction

- The Irish accent varies massively across the little island of Ireland and is a major part of the Irish identity.
- Using machine learning we can analyse and classify speech patterns ie. Accents.
- My project explores a system for predicting and mapping Irish accents from self-created datasets.

#### Project Aim

To design and implement a machine learning system for Irish accent classification and geospatial visualisation, supported by an experimental evaluation of model performance.

#### Key Findings

- Province-level classification performed poorly (~0.39 balanced accuracy).
- Binary classification tasks achieved significantly higher performance (~0.75–0.79).
- Accent differences are more detectable at broader regional levels.
- Model performance is strongly influenced by dataset quality and balance.

